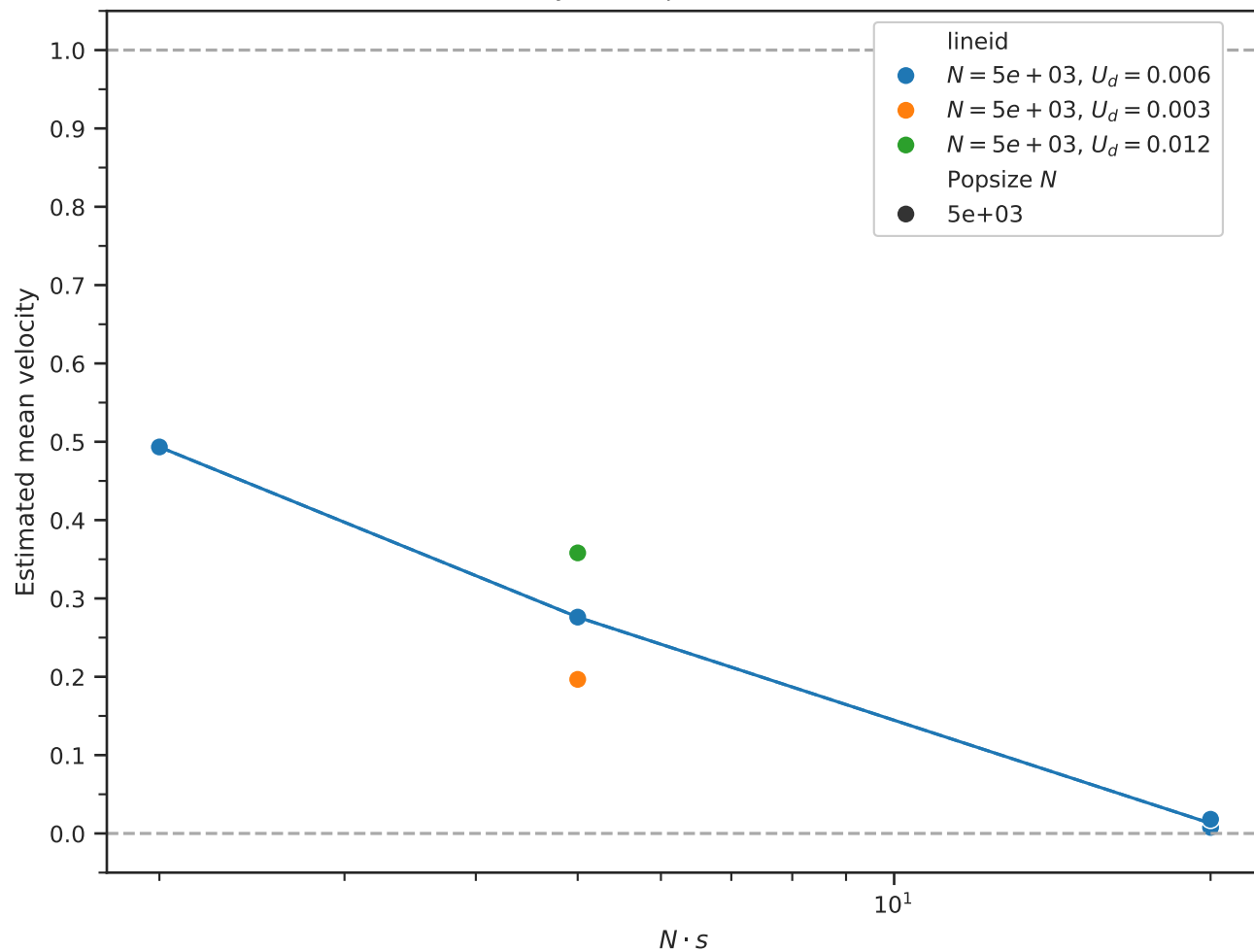
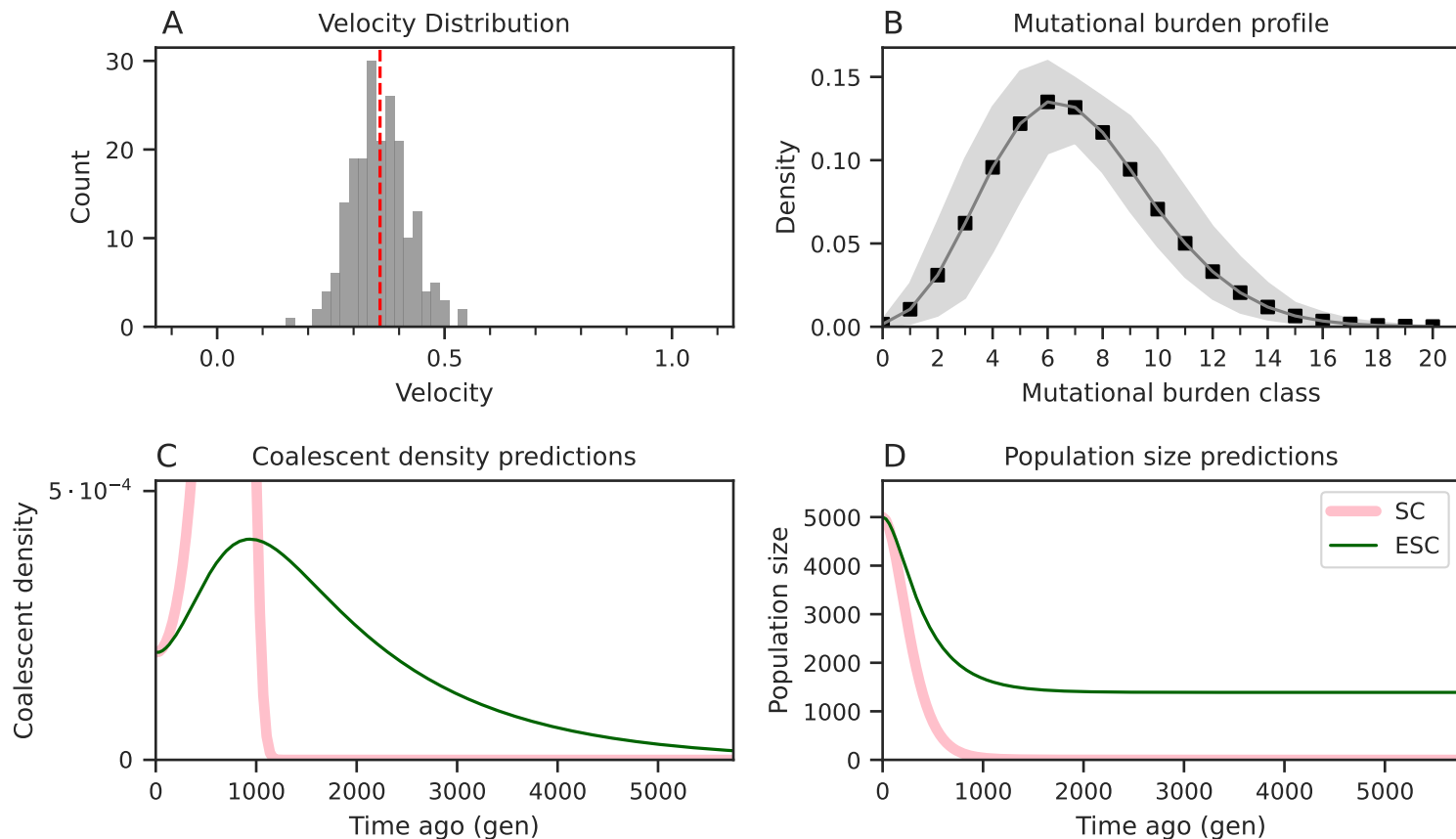


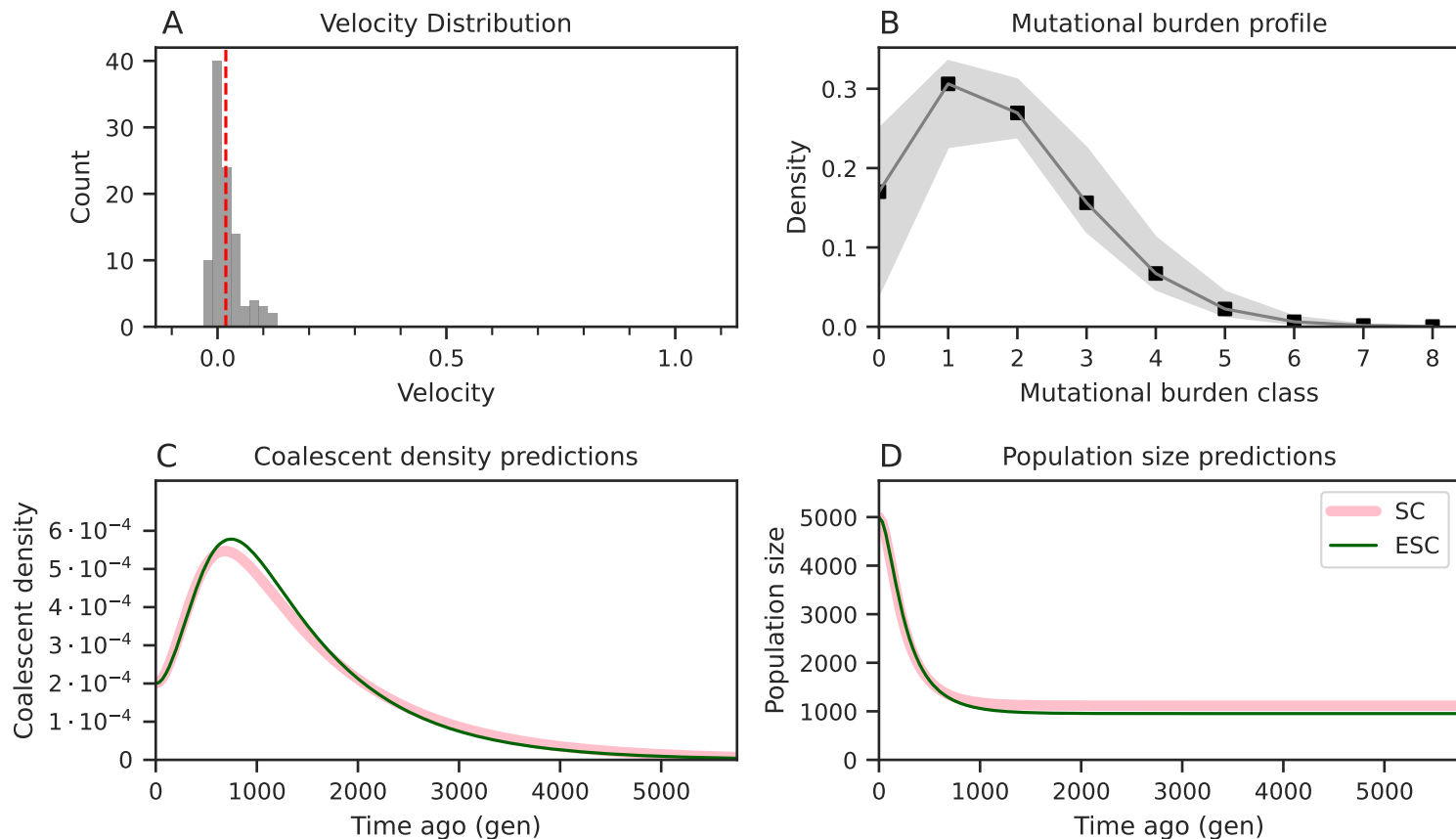
Mean velocity across parameter combinations



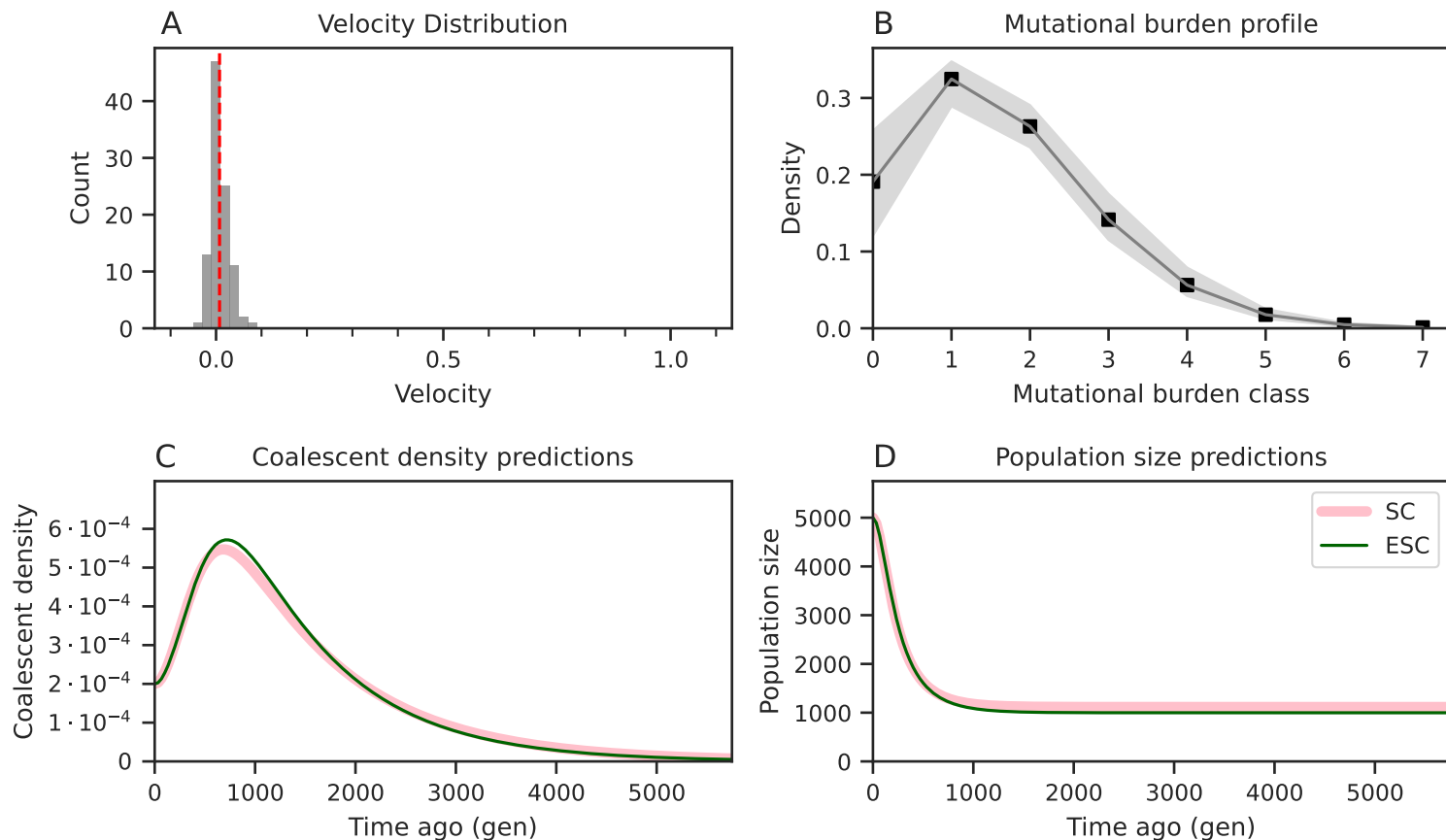
$N = 5e + 03$, $s = 0.001$, $U_d = 0.012$, $\sigma = 0$, $\phi = 3.07 \cdot 10^{-5}$
Mean Velocity= 0.3581 ± 0.0627 , $n = 200$ simulations



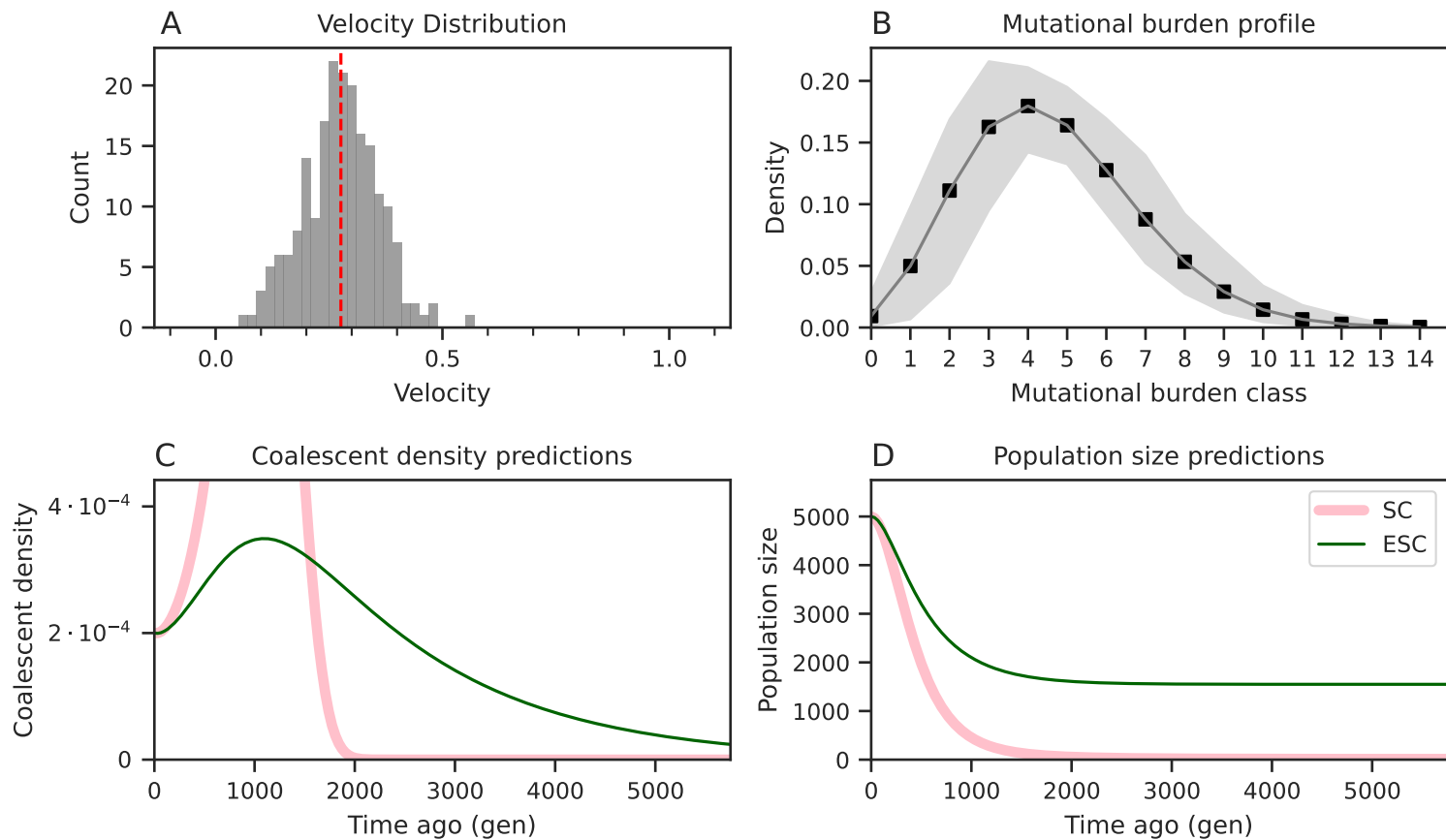
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 8.27 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0181 ± 0.0291 , $n = 100$ simulations



$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0$, $\phi = 4.46$
Mean Velocity= 0.0077 ± 0.0198 , $n = 100$ simulations

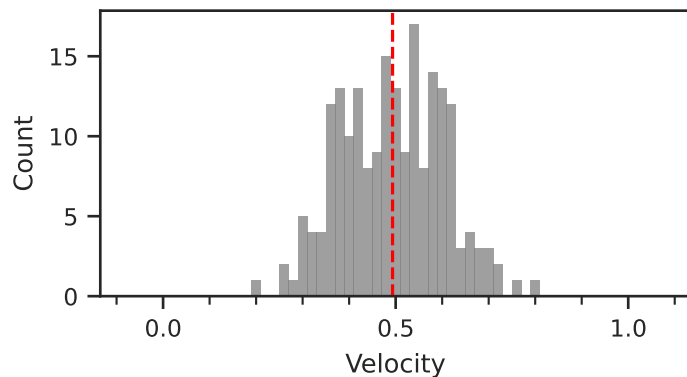


$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 0$, $\phi = 0.0124$
Mean Velocity = 0.2761 ± 0.0832 , $n = 200$ simulations

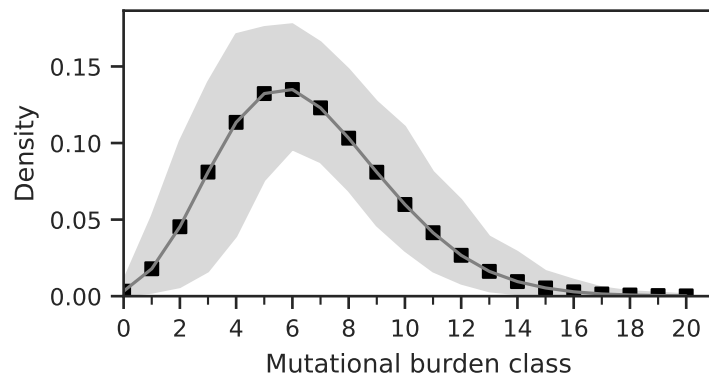


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 0$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.4933 ± 0.1093 , $n = 200$ simulations

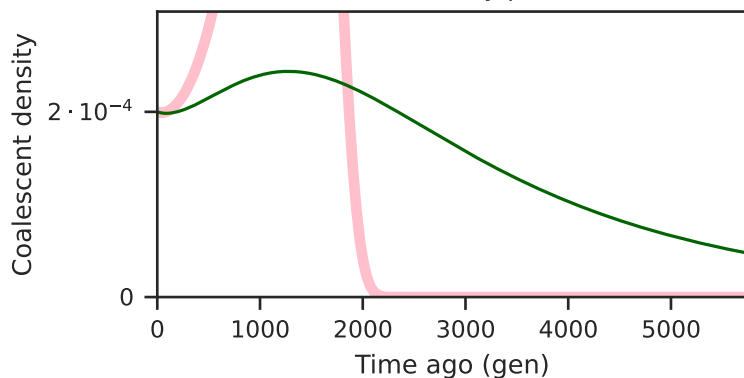
A Velocity Distribution



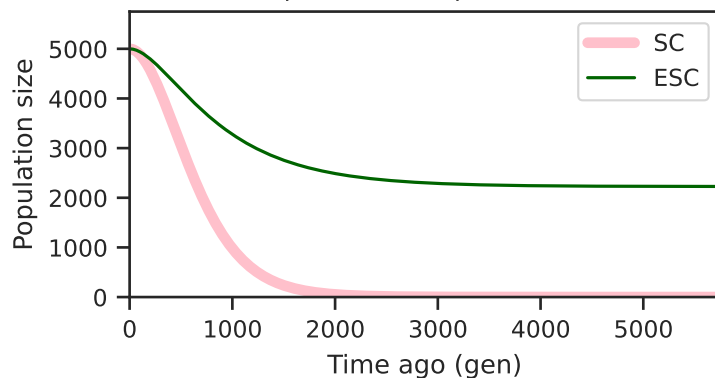
B Mutational burden profile



C Coalescent density predictions



D Population size predictions



$N = 5e + 03$, $s = 0.001$, $U_d = 0.003$, $\sigma = 0$, $\phi = 0.249$
Mean Velocity= 0.1967 ± 0.1112 , $n = 200$ simulations

